

VISHVESHWAR D S

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PROFILE

Electrical and Electronics Engineering graduate from Bangalore Institute of Technology with a strong focus on embedded systems and PCB design. Currently working as a Failure Analysis Engineer at Foxconn, specializing in hardware debugging, root cause analysis, and reliability assessment. Hands on experience with STM32, ESP32, and Arduino platforms, with strong proficiency in embedded C, firmware development, peripheral interfacing (I2C, SPI, UART), using STM32CubeIDE, CubeMX, and Arduino IDE. Skilled in schematic design, PCB layout, component selection, signal routing, and design validation using KiCad and Altium, with exposure to industry practices and design standards. Experienced in circuit simulation and analysis using MATLAB, Simulink, and LTspice. Completed embedded and sensor-based projects, including sensor fusion applications, demonstrating strong capabilities in system integration, testing, and optimization. Actively seeking opportunities to contribute to embedded hardware and PCB design domains.

PROFESSIONAL EXPERIENCE

2026/04 – Present
Bengaluru, India

FAE Engineer
FOXCONN INDIA PVT LTD

- Performed **failure analysis and root cause identification** on electronic hardware and PCB assemblies using systematic debugging techniques.
- Conducted **board-level troubleshooting**, including signal tracing, continuity checks, and component-level diagnosis.
- Utilized lab equipment such as **oscilloscopes, multimeters, and power supplies** for electrical testing and validation.
- Collaborated with cross-functional teams (design, manufacturing, quality) to implement **design improvements and process corrections**.
- Supported **reliability testing and validation** to ensure product performance and compliance with specifications.

SKILLS

Embedded C

I2C, SPI, UART protocols

PCB Design KICAD

MATLAB SIMULINK

LT Spice

EDUCATION

2020 – 2025
Bengaluru, India

B. E in Electrical and Electronics Engineering
Bangalore Institute Of Technology

PROJECTS

IOT based Li-Ion Battery Monitoring System using ESP8266 with Automatic Temperature control

- Developed a Battery temperature and capacity monitoring system and also implemented Automatic Temperature management using ESP8266 and LI-Ion battery.
- Configured the Arduino IOT Cloud and platform to publish the data via MQTT protocol.
- Improve the Battery cooling and temperature management achieved real time battery percentage and temperature remote monitoring using IOT and Arduino cloud platform.

PCB design of DC-DC Buck Converter using KICAD

- Designed a 2-layer Variable output DC-DC Buck Converter using KI-CAD, IC used is a LM2596 this circuit does not need a microcontroller.
- Max input voltage 40v, min output voltage is 1.23v, feedback voltage is 1.23v operating frequency is 150KHz.
- Without a help of a Microcontroller the feedback voltage of 1.23v is used to control the output duty cycle of the IC to enable variable input voltage added a voltage divider circuit using a 330ohm fixed resistor and a 10kohm variable pot so that the output voltage can be varied using the POT so that the output of the voltage divider is always 1.23v tricking the IC into bucking different input voltages lower than 40v .
- This buck converter can now reduce or buck the input dc voltage to a lower output voltage.

PCB design of an IR sensor module using KICAD

- Designed a PCB of an IR sensor module with adjustable sensitivity which is used as a proximity sensor.
- Consists of IR LED, IR receiver, LM358 module as main components. The LM358 IC acts as the comparator between fixed setup voltage and the variable voltage signal received from the IR receiver and also as the ADC to convert the compared output into digital signals for the microcontroller to process.
- Implemented variable sensitivity setting using a 10k ohm potentiometer to inverting pin and a voltage divider circuit consisting of IR receiver and 10k ohm resistor the non-inverting pin. The POT sets the threshold voltage by varying its resistance the vlg divider circuit, the reflected IR light on the receiver changes its resistance and the output of this divider circuit is compared with the set threshold voltage by the LM358 which then sends the output signal to the controller if the voltage is above the set voltage.

Design and fabrication of a Electric motor-powered E-bicycle with Regenerative braking

- Worked on the design and fabrication of an Electric motor-powered E-bicycle and also implemented Solar recharging
On the go along with regenerative braking.
- Implemented Regenerative braking on the rear brakes, when the rear brake pedals are pressed it brings

the generator wheel assembly closer to the rear wheel and starts spinning due to friction between the two wheels generating electricity which is then boosted by the MT3608 boost converter raising voltage from 6v to 24v to charge the battery.

- Implemented a Solar panel charging system which recharges the battery on the go or at rest at places where there is ample sunlight. Solar panel used is a 12v 10w solar panel which is boosted by a separate Boost converter to 24v to charge the battery.
- Max range 7 kms on full battery charge.

ADDITIONAL CERTIFICATIONS

- NPTEL From IIT Madras - Electrical Systems Design: Hands-On Circuits and PCB Design with CAD software.

NOV 2024

Skills: PCB Design KICAD, LT-Spice, Verilog Programming, Gtk wave analysis

- NIELIT Embedded for Beginners.

JAN 2026

Skills: Embedded C, STM32CUBEIDE SMART lab STM32G474 ARM MCU programming

- NPTEL PLUS Workshop: Building Better PCBs and Electronic Components: Thermal management using ANSYS ICEPAK.

Jul 2025

Skills: Thermal simulation and modelling of 8layer GPU PCB board in ICEPAK, Mesh creation, Thermal management

LANGUAGES

- English
- Kannada